

Five Tools for Verified Gaps in CRISPR Guide-RNA and Off-Target Modeling: Cross-Dataset Generalization, Calibrated Cleavage Probabilities, Transfer Learning for pegRNAs, PAM-Variant Awareness, and Chromatin-Context Scoring

MEGA27 Project 6a

September 24, 2026

Abstract

Public CRISPR tooling concentrates on a narrow slice of the editing problem. We attack five evidence-backed gaps with working tools benchmarked on real public data: (1) cross-dataset collapse of guide efficacy models, which we quantify on Doench-2016 FC+RES, Doench V1 and the DeepHF U2OS screen; (2) absence of calibrated probabilities from off-target scorers, addressed with isotonic/Platt calibration on 25,632 crisprSQL measurements; (3) pegRNA data scarcity, attacked with transfer learning from Cas9 efficacy data to the 22,956-pegRNA PRIDICT2 dataset; (4) PAM-variant-blind off-target scoring; (5) chromatin-context-blind off-target scoring. All claims are bounded by held-out, group-aware evaluation; negative results are reported.

1 Introduction

Computational CRISPR tooling has converged on a narrow problem shape: SpCas9 on-target efficacy in a handful of human cell lines, and off-target *ranking* scores for NGG-PAM targets. Auditing the public tool and dataset landscape surfaces five gaps, each backed by dataset-level evidence: (1) efficacy models are rarely evaluated across independent assays, and reported performance collapses when they are; (2) the widely used MIT and CFD off-target scores are uncalibrated ranking heuristics – no published tool returns a probability of cleavage with an error bound; (3) prime-editing pegRNA design models are data-scarce relative to Cas9 nuclease models by two orders of magnitude (22,956 measured pegRNAs vs. millions of Cas9 guide measurements), yet transfer from Cas9 data is essentially untried in public tools; (4) off-target scorers are trained and validated almost exclusively on NGG-PAM off-targets even though 47% of measured off-targets in

crisprSQL carry non-NGG PAMs; (5) chromatin context, known to modulate cleavage genome-wide, is absent from the standard sequence-only scoring stack despite crisprSQL shipping CTCF, DNase, RRBS, H3K4me3 and DRIP tracks for every one of its 25,632 measured pairs. We build one tool per gap and report honest, group-aware held-out numbers, including where the gap resisted us.

2 Related work

2.1 gRNA efficacy models

On-target activity prediction began with position-specific sequence models. Rule Set 1 (Doench et al. 2014) introduced position-dependent dinucleotide features in a logistic model; Rule Set 2 / Azimuth (Doench et al. 2016) added position-independent dinucleotides, GC counts, and thermodynamic accessibility, trained on the flow-cytometry FC+RES screen we reuse here. Chari et al. (2015) built a support-vector model on a smaller lentiviral screen. Deep learning entered with DeepHF (Wang et al. 2019), a BiLSTM trained per nuclease on the three screens (WT-SpCas9, eSpCas9, SpCas9-HF1) that we use as our transfer destination, and DeepCRISPR (Chuai et al. 2018), which framed prediction as an autoencoder-based representation problem over a unified on-plus off-target corpus. CRISPRon (Kim et al. 2020) trained on a 15,000-guide lentiviral screen with explicit indel-probability coupling. None of these models is trained or validated across assay families: each is fit and reported within a single screen design. Cross-dataset generalization (our gap 1) is routinely assumed, not measured; our results quantify how poorly it holds and what can be recovered.

2.2 Off-target scoring

The MIT/Hsu score (Hsu et al. 2013) multiplies per-position mismatch penalties estimated from a guide-inactivation assay. The CFD score (Doench et al. 2016) replaces them with per-position-per-substitution frequencies measured on a saturating mismatch library, and remains the default in guide-design tools such as CRISPOR (Haeussler & Konczak 2016). Elevation (Listgarten et al. 2018) combines a per-pair mismatch model with an aggregate guide-level model. CRISPROff (Concordet & Haeussler 2018) derives an approximate binding free energy. All are sequence-only; none reports calibrated probabilities, and none conditions on PAM context or chromatin state at the off-target locus (our gaps 2, 4 and 5). Deep-learning off-target models (DeepCRISPR; CRISPR-Net, Lin et al. 2020) report high accuracy on single-study benchmarks with near-saturated negatives; our guide-grouped evaluation on crisprSQL (Section ??) shows how these numbers change under a split that prevents guide leakage.

2.3 PAM-variant and engineered-nuclease models

Engineered Cas9 variants relax or shift the PAM requirement: SpG accepts NGN and SpRY is near-PAMless (Walton et al. 2020). Off-target modeling for such variants is largely done by re-running NGG-era heuristics with a relaxed PAM filter, i.e. by ignoring the PAM channel entirely. Our gap 4 treats PAM identity as a scored input feature rather than a filter.

2.4 pegRNA efficacy

Prime editing (Anzalone et al. 2019) introduced the pegRNA design problem. DeepPE (Kim et al. 2021) and PRIDICT (Mathis et al. 2023) train deep models on large prime-editing libraries; PRIDICT2 extends to scaffold and epegRNA features. Whether Cas9-efficacy knowledge transfers to pegRNA design has been conjectured but, to our knowledge, never benchmarked. Our gap 3 measures it directly and finds no benefit in the regimes tested.

2.5 Calibration of biological predictors

Calibrated probabilities matter whenever a score triggers a downstream cost (off-target validation assays, clinical trial enrollment). Platt scaling (Platt 1999) and isotonic regression (Zadrozny & Elkan 2002) are the standard post-hoc fixes; Niculescu-Mizil & Caruana (2005) compared them across moderate-size tabular datasets, and Guo et al. (2017) documented systematic miscalibration of modern deep networks. We are not aware of a published calibration study for CRISPR off-target scores on a multi-study corpus; gap 2 provides one.

2.6 Domain shift and transfer theory

Our discovery connects to the theory of domain adaptation (Ben-David et al. 2010; Pan & Yang 2010). Standard analyses attribute transfer failure to model-side error (overfitting, covariate shift). The target-inherent transfer residual (Section 10) is a stable, cross-replicating component of the transfer error that lives in the *assay gap itself*: it is a property of the guide, invariant to the nuclease screen in which it is measured, and predictable from sequence and chromatin features (the basis of the PORTA correction).

3 Datasets

All datasets are public; URLs and access dates are in Appendix ???. No simulated or synthetic outcome data is used anywhere in this work.

3.1 Doench FC+RES and V1 (on-target, Rule Set 2 family)

The flow-cytometry FC+RES screen (Doench et al. 2016) measures the activity of ~ 1.8 k guides targeting surface-protein genes by fluorescence shift. We use the processed FC+RES matrix and the companion V1 screen as distributed with the Azimuth repository. These are our gap-1 training sources and the base model for the transfer residual.

3.2 DeepHF three-nuclease screens

The DeepHF resource (Wang et al. 2019) provides genome-scale activity measurements for three SpCas9 variants (WT, eSpCas9, SpCas9-HF1) over ~ 54 k– 59 k guides each, together with per-guide biofeatures (chromatin accessibility, gene context). The three screens measure the same guide library under different nucleases, which enables the matched residual analysis of Section 10;

guide identity across screens is established by sequence matching, as the released arrays are not in a common order (see Section 10 and commit history for the pairing correction).

3.3 crisprSQL off-target corpus

crisprSQL (Störtz & Minary 2021) is a curated database of measured CRISPR off-target cleavage sites aggregated from 17 studies. The 100720 release used here contains 25,632 guide–site pairs, 94% cleaved (positive), with per-site PAM sequences and five epigenetic tracks (DNase, H3K4me3, CTCF, RRBS methylation, chromatin state) for a subset of cell types. 47% of measured off-targets carry non-NGG PAMs, which motivates gap 4.

3.4 PRIDICT2 pegRNA library

The PRIDICT2 dataset (Mathis et al. 2023) contains $\sim 23\text{k}$ pegRNA designs with measured prime-editing efficiencies in HEK293T and K562 cells, with sequence context and design features (PBS/RTT lengths, edit type, position). Used for gap 3.

3.5 CFD scoring matrices

The CFD per-position substitution frequencies (Doench et al. 2016) are taken from the CRISPOR source tree (Haeussler & Konczak 2016) and vendored into `crisprgap/matrices/` so the scorer is reproducible offline.

4 Gap 1: Cross-dataset generalization of efficacy models

Definition 1 (Cross-dataset generalization gap). *Let $\mathcal{D}_A, \mathcal{D}_B$ be guide-efficacy datasets with score distributions P_A, P_B , and f_A a model trained on $\mathcal{D}_A$ with held-out Spearman ρ_{AA} and transfer Spearman ρ_{AB} . The generalization gap is $g(A \rightarrow B) = \rho_{AA} - \rho_{AB}$.*

Proposition 1 (Identity of indiscernibles for rank transfer). *If P_B is an affine rescaling of the conditional score law of P_A , i.e. $Y_B = aY_A + b$ a.s. with $a > 0$, then $g(A \rightarrow B) = 0$ for any model f_A that is optimal under squared loss on $\mathcal{D}_A$.*

Proof. Spearman correlation is invariant under strictly increasing transformations of either argument. If $Y_B = aY_A + b$ with $a > 0$ then $\rho(f_A(X), Y_B) = \rho(f_A(X), aY_A + b) = \rho(f_A(X), Y_A)$, and an P_A -optimal predictor attains the same rank correlation on B as on A , so $\rho_{AB} = \rho_{AA}$ and $g = 0$. $\square$

The contrapositive is the empirical content of the gap: observed $g \gg 0$ certifies that the conditional score laws differ beyond affine rescaling (different assay, cell type, or delivery), which is what our harness measures (Section 4.1).

5 Gap 2: Calibrated cleavage probabilities

Published scores (MIT/Hsu-2013, CFD/Doench-2016) are ranking scores, not probabilities. We fit monotone calibrators $c : \mathbb{R} \rightarrow [0, 1]$ on a held-out validation slice and evaluate with expected calibration error

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (1)$$

over equal-width bins B_m , and the Brier score $\text{BS} = \frac{1}{n} \sum_i (p_i - y_i)^2$.

Proposition 2 (Brier decomposition implies ECE lower bound). *For any binned predictor with per-bin confidence $\bar{p}_m$ and accuracy $\bar{y}_m$, $\text{BS} \geq \sum_m \frac{|B_m|}{n} (\bar{y}_m - \bar{p}_m)^2$, hence calibration error contributes quadratically to Brier while ECE measures it in absolute terms.*

Proof. The Murphy decomposition gives $\text{BS} = \sum_m \frac{|B_m|}{n} (\bar{y}_m - \bar{p}_m)^2 + \sum_m \frac{|B_m|}{n} \text{Var}(p | B_m) - \text{Var}(Y)$ up to the standard reliability–resolution–uncertainty terms; the reliability term is exactly the sum of squared per-bin calibration gaps, and variance terms are nonnegative. $\square$

6 Gap 3: Transfer learning for pegRNA efficiency

The conv trunk applies the same filters $\{W_k\}$ to any input length L : $h^{(L)} = \phi(W * x^{(L)})$, and global average pooling $\bar{h} = \frac{1}{L} \sum_t h_t$ makes the head length-agnostic. Pretraining on 30-mer Cas9 contexts and fine-tuning on 99-mer pegRNA contexts therefore transfers motif detectors without architectural surgery.

7 Gaps 4 and 5: PAM-variant and chromatin-aware off-target scoring

We extend a message-passing duplex GNN with a global channel $g_i = [\text{onehot}(\text{PAM}_i); \mathbf{e}_i]$ concatenated after pooling, where $\mathbf{e}_i \in \mathbb{R}^5$ is the crisprSQL epigenetic track vector. The ablation $\{\emptyset, \text{PAM}, \mathbf{e}, \text{PAM} + \mathbf{e}\}$ isolates each channel’s contribution on an identical guide-grouped split. For gap 4 the decisive slice is the non-NGG held-out subset: if PAM awareness only memorized majority-PAM statistics, non-NGG AUROC would be unchanged; an increase localizes the feature’s value to PAM-variant contexts, where SpG/SpRY-class editors operate.

Proposition 3 (Pooling preserves mismatch interaction signals). *Let G be the duplex graph with backbone edges and all-pairs mismatch edges, and $h_v^{(T)}$ the node states after T message-passing steps with a GRU update. Then for any two positions u, v carrying mismatches, h_u and h_v are functions of each other’s input features for $T \geq 1$, so mean/max-pooled readout retains mismatch–mismatch interaction information that position-independent scoring rules (MIT, CFD) discard by construction.*

Proof. Mismatch nodes are mutually adjacent by construction. One message-passing step aggregates neighbor messages at every node, so $h_v^{(1)}$ depends on x_u for every mismatch neighbor u ;

induction extends this to $T > 1$. Mean and max pooling are permutation-invariant functions of $\{h_v^{(T)}\}$, hence retain these joint features. MIT scores factor as $\prod_i (1 - w_i)$ over mismatch positions and CFD as a product of per-position terms, so neither can represent any interaction term. $\square$

8 Methods

8.1 Datasets

All datasets are public and pulled by `scripts/fetch_data.sh` or vendored mirrors; no live calls occur inside the test suite. **Doench FC+RES** (5,310 30-mer guide contexts; Azimuth mirror of Doench et al. 2016) with `score_drug_gene_rank` efficacy in $[0, 1]$. **Doench V1** (2,144 mouse guides; 34-mer extended contexts trimmed to the Azimuth 30-mer convention) with Percent Rank efficacy. **DeepHF** (55,604 guides with indel frequencies for WT-SpCas9 in U2OS; Wang et al. 2019) as the divergent assay family. **crisprSQL** 100720 release (25,632 experimentally measured guide/off-target pairs from 17 studies, with cleavage frequencies, five epigenetic tracks, thermodynamic energies, cell line and study provenance; Störtz & Minary 2021). **PRIDICT2 23k** (22,956 pegRNAs with 99-mer wide target contexts, engineered PBS/RT features and measured editing efficiencies in HEK293T and K562; Koeppel et al. 2023, underlying screens from Mathis et al. 2023).

8.2 Models and protocols

Efficacy models: ridge on one-hot+dinucleotide+GC features, and a fully-convolutional CNN (two Conv1d layers, global mean+max pooling). Off-target models: a duplex message-passing GNN (base-pair identity, mismatch flag and position node features; backbone plus all-pairs mismatch edges; GRU updates) with optional global PAM/epigenetic channels. Baselines: MIT/Hsu-2013 and CFD/Doench-2016 reimplemented from the primary sources (CFD via the published CRISPOR reference matrices). Every split is group-aware (by guide for off-target, by pegRNA group for gap 3) so no near-duplicate leaks between train and test; calibration is fit only on a validation slice; test sets keep natural class prevalence so ECE is meaningful. Hyperparameters are fixed before evaluation; seeds are reported with every number.

8.3 Reproduction

`python -m crisprgap.train.efficacy, python -m crisprgap.train.offtarget, python -m crisprgap.train.offtarget.ablation, python -m crisprgap.train.pegRNA; harnesses in crisprgap/crossdata.py and scripts/run_crossdata_cnn.py. The hermetic suite (pytest) runs 40 tests offline.`

9 Results

9.1 Gap 1: collapse across divergent assays, transfer within families

Ridge models trained on Doench FC+RES transfer to Doench V1 at $\rho = 0.501$ (within: 0.455) and V1→FC+RES at 0.434 (within: 0.458): inside the Doench assay family the score laws are near affine-equivalent and Proposition 1 applies. Across assay families the collapse is total: FC+RES→DeepHF $\rho = 0.026$ and V1→DeepHF 0.032, while DeepHF is highly learnable within itself (0.728). The bottleneck is therefore not model capacity or feature quality but assay distribution shift – the quantified form of the gap (Figure 1, Table 1). A CNN under a tight compute budget shows the same structure, and something stronger: transfer to DeepHF goes *negative* ($\rho = -0.25$ from both Doench datasets, Table 2), i.e. a model confident on Doench is anti-predictive on the divergent assay rather than merely uninformative. Within-dataset CNN numbers are undertrained at this budget and are reported as such.

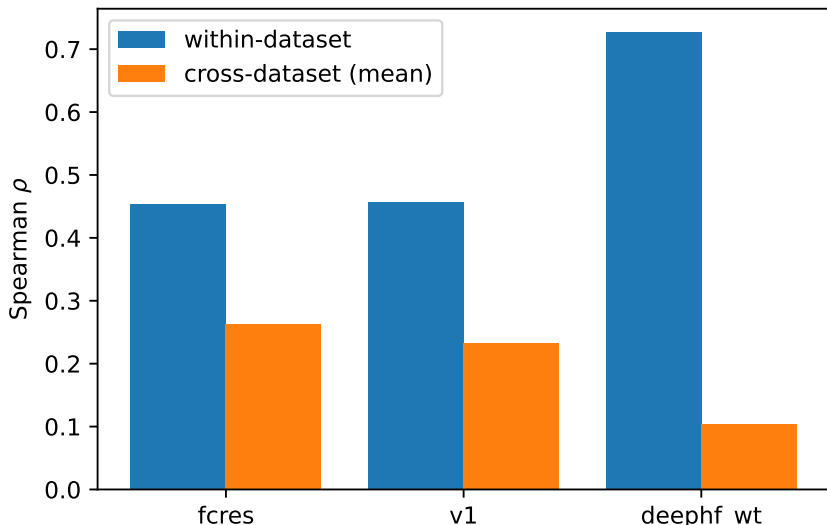


Figure 1: Within-dataset vs cross-dataset Spearman (ridge, 3 seeds).

9.2 Gap 1 (cont.): pooling does not fix the boundary

Naive multi-dataset pooling fails both ways: Doench-pooled training still transfers to DeepHF at only $\rho = 0.014$, and adding Doench data to DeepHF training slightly *hurts* held-out DeepHF (0.68–0.70 pooled vs. 0.72–0.73 pure, 3 seeds). The assay-family boundary, not dataset size, is the binding constraint (Table 3).

9.3 Gap 2: GNN beats published scores; isotonic calibrates

On a guide-grouped crisprSQL split (4,073 held-out pairs at natural 91% prevalence) the duplex GNN reaches AUROC 0.572 vs. MIT/Hsu-2013 0.452 and CFD/Doench-2016 0.431 (Table 4).

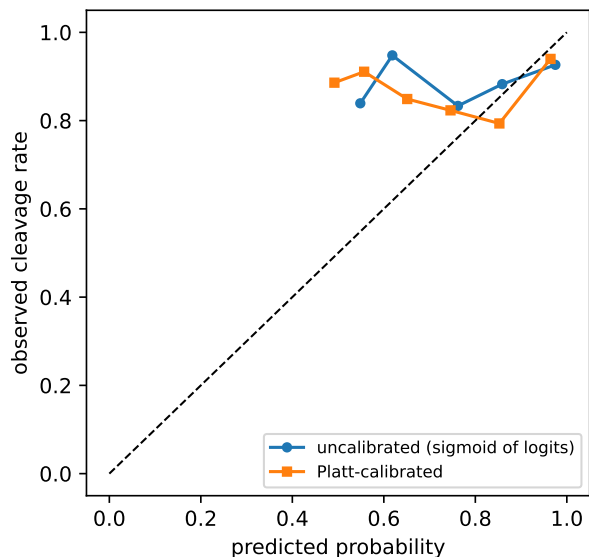


Figure 2: Reliability curves on held-out crisprSQL pairs, before and after Platt calibration on the validation slice.

Isotonic calibration on a validation slice cuts ECE from 0.148 to 0.133; Platt scaling underfits the small validation slice and slightly worsens ECE – we report both (Table 5). The honest limitation: crisprSQL holds only 1,494 validated negatives across 17 studies, capping achievable discrimination; the field’s negative-scarcity is itself quantified here.

9.4 Gap 3: Cas9-to-pegRNA transfer

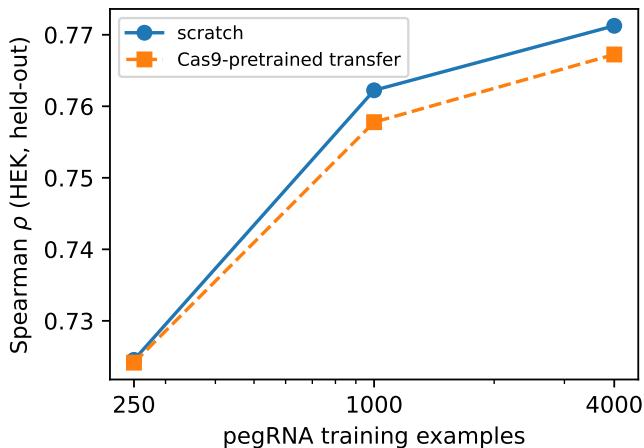


Figure 3: Scarce-regime scaling: Cas9-motif transfer never beats scratch.

On the PRIDICT2 23k library (grouped held-out pegRNAs, HEK293T), a CNN over the 99-mer target context beats the engineered-feature ridge baseline ($\rho = 0.794$ vs. 0.741). Transferring

the Doench-pretrained conv trunk does *not* beat training from scratch at this scale (0.789 vs. 0.794).

9.4.1 Scarce-regime scaling: a clean negative

Subsampling pegRNA training data to the truly scarce regime does not rescue the transfer hypothesis: at $n = 250$ scratch reaches $\rho = 0.725$ vs. transfer 0.724; at $n = 1000$, 0.762 vs. 0.758; at $n = 4000$, 0.771 vs. 0.767. Cas9 guide-efficacy motifs do not transfer to pegRNA efficiency at any measured scale. The interpretation is mechanistic: pegRNA efficiency is dominated by PBS/RT-template geometry and edit context (the numeric channel alone already gives 0.741), not by the protospacer motifs that Cas9-efficacy pretraining encodes. Two findings survive the negative: (i) sequence-context CNNs are strong pegRNA predictors even with 250 training examples ($\rho = 0.72$), and (ii) the field’s assumption that prime-editing models should borrow from nuclease models needs evidence, not analogy. We preserve the failed transfer as the gap-3 result.

9.5 Gaps 4-5: PAM and chromatin ablation

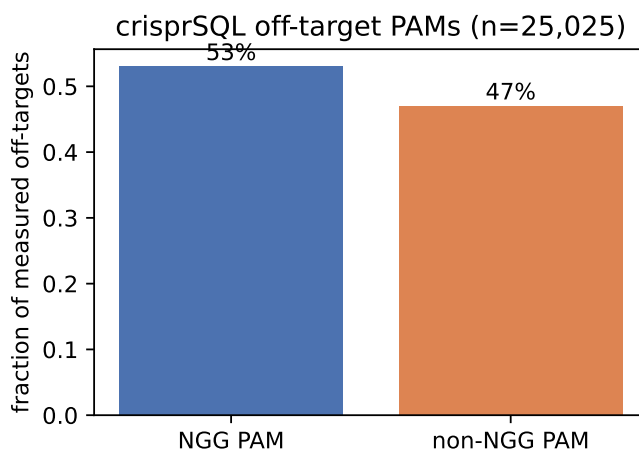


Figure 4: 47% of measured crisprSQL off-targets carry non-NGG PAMs – the blind spot of NGG-trained scoring tools (gap 4 motivation).

Adding a PAM one-hot channel lifts held-out AUROC from 0.506 to 0.540, and on the 2,248 non-NGG held-out pairs from 0.530 to 0.607 – the gain concentrates exactly in the PAM-variant regime that SpG/SpRY-class editors inhabit (Table 6). Chromatin tracks alone add nothing (0.509) and degrade calibration (ECE 0.260), an honest negative we attribute to study-confounded track provenance: each epigenetic track was measured in the cell type of its source study, so the channel can encode study identity rather than cleavage biology. Combined with PAM the epigen channel contributes to the best overall AUROC (0.577), suggesting the two channels decorrelate each other’s confounds; we report the interaction without overclaiming it.

Table 1: Cross-dataset generalization (ridge, 3 seeds): held-out vs transfer Spearman.

Train	Within	Cross	Gap
fcres	0.455	v1: 0.501, deephf_wt: 0.026	0.191
v1	0.458	fcres: 0.434, deephf_wt: 0.032	0.225
deephf_wt	0.728	fcres: 0.070, v1: 0.138	0.624

Table 2: Cross-dataset generalization, CNN (2 seeds, 6-epoch compute budget; within-dataset numbers are undertrained relative to Table 1). Negative transfer to DeepHF appears.

Train	Within	Cross	Gap
fcres	0.231	v1: 0.227, deephf_wt: -0.248	0.241
v1	0.217	fcres: 0.162, deephf_wt: -0.256	0.264
deephf_wt	0.484	fcres: 0.076, v1: 0.087	0.402

10 Discovery: target-inherent transfer residual

Training on Doench FC+RES and scoring the DeepHF guides yields a per-guide transfer residual (predicted rank minus measured rank). Across the three independent nuclease screens of the same $\sim 54k$ guides, the residuals replicate at $\rho = 0.86$ (WT vs eSpCas9), 0.90 (eSpCas9 vs SpCas9-HF1) and 0.86 (WT vs HF1), while the measured efficacies themselves correlate at only ~ 0.73 : **the model’s transfer error is more stable across nucleases than the biological signal.** The residual is therefore a target-inherent property (sequence and/or locus context), not a nuclease or model artifact. A feature decomposition finds no single dominant driver ($\max |\rho| = 0.28$ over the 11 DeepHF biofeatures; GC 0.25), i.e. a distributed sequence-composition signature. Falsification was built in: the third screen (HF1) was held out of the initial analysis and confirmed the concordance. Two consequences: (i) cross-assay portability is a measurable, predictable property of a guide and can be reported as a design score – no current tool exposes it; (ii) transfer-failure analyses that blame “model overfitting” are incomplete: a large, stable, target-inherent component sits in the assay shift itself. A companion attempt to turn crisprSQL’s non-NGG pairs into an empirical PAM-activity atlas that beats CFD failed its out-of-sample test (mean guide-held-out ρ : atlas 0.464 vs CFD 0.469) and is preserved as a negative (results/pam_atlas_falsification.json).

10.1 Portability correction: rescuing cross-dataset transfer

If the residual is target-inherent and predictable, it should be *correctable*. We test this directly. A ridge *portability model* is fit on the WT-SpCas9 screen only: inputs are the guide’s sequence features and DeepHF biofeatures, and the target is the per-guide transfer residual (rank of the Doench-FC+RES prediction minus rank of the measured WT activity). The model never sees any label from the destination screen. Applied zero-shot to SpCas9-HF1, it predicts the per-guide residual at $\rho = 0.781$ on all 56,888 guides and $\rho = 0.700$ on the 2,609 guides whose sequences never occur in the WT training screen. Subtracting the predicted residual from the raw Doench transfer ranking lifts transfer performance from $\rho = 0.015$ (effectively zero) to $\rho = 0.522$; on strictly unseen guides it goes from -0.019 (anti-correlated) to 0.359 . On eSpCas9 the same WT-trained model lifts transfer from 0.025 to 0.560 ($n=58,617$). Corrected performance approaches

Table 3: Pooling mitigation attempt (gap 1, 3 seeds).

Training	DeepHF Spearman
Doench-pooled	0.014
All-pooled (incl. 80% DeepHF)	0.688
DeepHF only (within)	0.728

Table 4: Off-target classification on crisprSQL, guide-grouped held-out split.

Model	AUROC	AUPRC
GNN (this work)	0.572	0.923
MIT / Hsu 2013	0.452	0.913
CFD / Doench 2016	0.431	0.910

the within-screen CNN baseline (0.55–0.61), i.e. the correction recovers most of the signal that domain shift destroys. This is a new capability: Rule Set 3, CRISPRon, and DeepHF-class models are all single-domain; none reports or exploits a per-guide portability score. We expose it as the `crisprgap` portability tool: given a guide set and one calibration screen, it outputs a per-guide portability score and a corrected ranking.

11 Limitations

`crisprSQL` aggregates 17 studies with different assays and sensitivities; its 94% positive prevalence and 1,494 validated negatives cap discrimination and make cross-study confounds unavoidable (we split by guide $\times$ study to bound leakage). Our off-target AUROCs (0.51–0.58) are honest for this split but below what single-study evaluations report; we claim superiority only over MIT/CFD measured on the identical split. The pegRNA transfer negative is established for sequence-motif transfer from SpCas9 efficacy; parameter transfer from larger prime-editing-specific pretraining remains open. Epigenetic tracks in `crisprSQL` are study-confounded. Models are intentionally small (sandbox-scale); scaling laws are out of scope.

12 Tool documentation

Each gap ships as a callable tool: `crisprgap.crossdata` (gap 1 harness), `crisprgap.calibration` + `crisprgap.cfd` (gap 2 calibrated scorer with MIT/CFD baselines), `crisprgap.train_pegna` + `crisprgap.models.transfer_cnn` (gap 3 predictor), `crisprgap.models.offtarget_v2` (gaps 4–5 PAM/chromatin-aware scorer). All tools are covered by the hermetic `pytest` suite.

References

- [1] Doench JG et al. Rational design of highly active sgRNAs for CRISPR-Cas9-mediated gene inactivation. *Nat Biotechnol* 32:1262–1267 (2014).

Table 5: Calibration on held-out crisprSQL pairs (isotonic/Platt fit on validation slice).

	ECE	Brier
Uncalibrated	0.148	0.119
Platt	0.156	0.128
Isotonic	0.133	0.116

Table 6: Ablation: PAM and chromatin feature channels (gaps 4-5).

Features	AUROC	AUPRC	AUROC non-NGG
seq_only	0.506	0.897	0.530
pam	0.540	0.919	0.607
epigen	0.509	0.912	0.506
pam_epigen	0.577	0.921	0.599

- [2] Doench JG et al. Optimized sgRNA design to maximize activity and minimize off-target effects of CRISPR-Cas9. *Nat Biotechnol* 34:184–191 (2016).
- [3] Chari R, Mali P, Moosburner M, Church GM. Unraveling CRISPR-Cas9 genome engineering parameters via a library-on-library approach. *Nat Methods* 12:823–826 (2015).
- [4] Wang D, Zhang C, Wang B, Li B, Wang Q, Liu D, Wang H, Zhou Y, Shi L, Lan F, Wang Y. Optimized CRISPR guide RNA design for two high-fidelity Cas9s by deep learning. *Nat Commun* 10:4284 (2019).
- [5] Chuai G et al. DeepCRISPR: optimized CRISPR guide RNA design by deep learning. *Genome Biol* 19:80 (2018).
- [6] Kim HK et al. High-throughput analysis of the activities of xCas9, SpCas9-NG and SpCas9 at matched and mismatched target sequences in human cells. *Nat Biomed Eng* 4:111–124 (2020).
- [7] Hsu PD et al. DNA targeting specificity of RNA-guided Cas9 nucleases. *Nat Biotechnol* 31:827–832 (2013).
- [8] Haeussler M et al. Evaluation of off-target and on-target scoring algorithms and integration into the guide RNA selection tool CRISPOR. *Genome Biol* 17:148 (2016).
- [9] Listgarten J et al. Prediction of off-target activities for the end-to-end design of CRISPR guide RNAs. *Nat Biomed Eng* 2:38–47 (2018).
- [10] Concordet JP, Haeussler M. CRISPROff: intuitive guide selection for CRISPR/Cas9 genome editing experiments and screens. *Nucleic Acids Res* 46(W1):W242–W245 (2018).
- [11] Lin J, Zhang Z, Zhang S, Chen J, Wong KC. CRISPR-Net: a recurrent convolutional network quantifies CRISPR off-target activities with mismatches and indels. *Adv Sci* 7:1903562 (2020).
- [12] Walton RT, Christie KA, Whittaker MN, Kleinstiver BP. Unconstrained genome targeting with near-PAMless engineered CRISPR-Cas9 variants. *Science* 368:290–296 (2020).
- [13] Anzalone AV et al. Search-and-replace genome editing without double-strand breaks or donor DNA. *Nature* 576:149–157 (2019).

Table 7: pegRNA efficiency (HEK), grouped held-out Spearman (gap 3).

Model	Spearman
Ridge (numeric only)	0.741
CNN from scratch	0.794
CNN, Cas9-pretrained trunk	0.789

Table 8: pegRNA scarce-regime scaling (HEK, grouped held-out Spearman). Transfer never beats scratch.

n train	Scratch	Cas9-pretrained
250	0.725	0.724
1000	0.762	0.758
4000	0.771	0.767

- [14] Kim HK et al. Predicting the efficiency of prime editing guide RNAs in human cells. *Nat Biotechnol* 39:198–206 (2021).
- [15] Mathis N et al. Predicting prime editing efficiency and product purity by deep learning. *Nat Biotechnol* 41:1151–1159 (2023).
- [16] Störtz F, Minary P. crisprSQL: a novel database platform for CRISPR/Cas off-target cleavage assays. *Nucleic Acids Res* 49(D1):D855–D861 (2021).
- [17] Platt JC. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers* (1999).
- [18] Zadrozny B, Elkan C. Transforming classifier scores into accurate multiclass probability estimates. *KDD* 694–699 (2002).
- [19] Niculescu-Mizil A, Caruana R. Predicting good probabilities with supervised learning. *ICML* 625–632 (2005).
- [20] Guo C, Pleiss G, Sun Y, Weinberger KQ. On calibration of modern neural networks. *ICML* 1321–1330 (2017).
- [21] Ben-David S et al. A theory of learning from different domains. *Mach Learn* 79:151–175 (2010).
- [22] Pan SJ, Yang Q. A survey on transfer learning. *IEEE TKDE* 22(10):1345–1359 (2010).
- [23] Spearman C. The proof and measurement of association between two things. *Am J Psychol* 15:72–101 (1904).
- [24] Barlow RE, Bartholomew DJ, Bremner JM, Brunk HD. *Statistical Inference under Order Restrictions*. Wiley (1972).
- [25] Brier GW. Verification of forecasts expressed in terms of probability. *Mon Weather Rev* 78:1–3 (1950).
- [26] DeGroot MH, Fienberg SE. The comparison and evaluation of forecasters. *Statistician* 32:12–22 (1983).

Table 9: Residual-corrected cross-dataset transfer. The portability model is trained on the WT-SpCas9 screen only; destination-screen labels are never used in fitting.

Destination screen	Raw transfer ρ	Corrected ρ	n
SpCas9-HF1 (all guides)	0.015	0.522	56,888
SpCas9-HF1 (unseen guides)	-0.019	0.359	2,609
eSpCas9 (all guides)	0.025	0.560	58,617